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A Reliability Generalization Study on the Survey of Perceived Organizational Support

The Effects of Mean Age and Number of Items on Score Reliability

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The Survey of Perceived Organizational Support (SPOS) is a unidimensional measure of the general belief held by an employee that the organization is committed to him or her, values his or her continued membership, and is generally concerned about the employee's well-being. In the interest of efficiency, researchers are often compelled to use a minimum number of SPOS items in their studies. This study reports on a reliability generalization across 62 published studies using the SPOS. Findings suggest that number of SPOS items and mean age of the sample are statistically significant in the relationship to reliability estimates. Additionally, mean age accounted for significant variance in internal consistency estimates over and above the number of items used.

Keywords: *perceived organizational support; reliability generalization; mean age*

Employers and employees exist in a system of mutual dependence (cf. Gouldner, 1960). Employees provide specific workplace services and in return expect rewards from the employer (e.g., pay, recognition). Likewise, organizations provide a reward structure and expect employees to be loyal and productive. Eisenberger, Huntington, Hutchison, and Sowa (1986) suggested that employees form general beliefs based on their perception that the organization is committed to them, values

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their continued membership, and is generally concerned about their well-being. They named this belief perceived organizational support (POS). POS is grounded in social exchange theory (Blau, 1964) and the norm of reciprocity (Gouldner, 1960), suggesting that employees give special attention to the effort of the organization to recognize and reward their work place behaviors. To the extent that treatment by the organization is viewed as favorable, the employee develops a sense of obligation to engage in behaviors that help the organization succeed (Eisenberger, Jones, Aselage, & Sucharski, 2004).

For the past two decades, POS has received promising empirical support, generating increased attention by both researchers and practitioners, thus advancing the understanding of organizational behavior. Research in applied settings has found POS to be positively related to performance of conventional job responsibilities, citizenship behaviors, and commitment (Eisenberger et al., 1986; Eisenbeger, Fasolo, & Davis-LaMastro, 1990; Shore & Wayne, 1993). Furthermore, POS scores have been shown to correlate with the hypothesized antecedent and consequent constructs providing further support for the theoretical relevance of organizational support. Indeed, a recent meta-analysis of 70 empirical studies (Rhoades & Eisenberger, 2002) showed that POS correlates meaningfully with perceived fairness and supervisor support (Nye & Witt, 1993) along with important outcomes such as job satisfaction (Witt, 1991), organizational commitment (Dawson, 1996), performance (Daly, 1998), and withdrawal behaviors (Allen, Shore, & Griffeth, 2003; Aquino & Griffeth, 1999).

The Survey of Perceived Organizational Support (SPOS)

The SPOS was originally developed as a 36-item self-report measure presented with a 7-point disagree-agree Likert-type response format. Shorter 16-item and 8-item versions of the original SPOS have also been used. An exploratory factor analysis of the original 36-item SPOS scale (Eisenberger et al., 1986) and subsequent exploratory and confirmatory analyses of shorter versions of the SPOS measure provide evidence for unidimensionality, reliability, and acceptable item-total correlations (Armeli, Eisenberger, Fasolo, & Lynch, 1998; Eisenberger et al., 1986; Hutchison, 1997; Shore & Tetrick, 1991). Likewise, there is evidence that SPOS scores were distinct in comparison with other measures such as job satisfaction (Aquino & Griffeth, 1999), perceived supervisor support (Kottke & Sharafinski, 1988), and affective commitment (Eisenberger et al., 1990; Rhoades, Eisenberger, & Armeli, 2001; Settoon, Bennett, & Liden, 1996; Shore & Tetrick, 1991). Evidence from one confirmatory factor analysis, however, suggested that POS and perceived organizational politics perhaps overlap (Randall, Cropanzano, Bormann, & Birjulin, 1999), thus emphasizing the need for further discriminant validity.

Purpose of the Study

Given that the SPOS is a widely used measure in organizational research, evaluating the psychometric properties of its scores could increase the level of confidence in interpreting empirically based conclusions. The purpose of this study is to present a reliability generalization (RG) analysis across studies using the SPOS as formulated by Eisenberger et al. (1986). Whereas the theoretical and empirical support for the construct of POS is promising, research surrounding *score* reliability of the SPOS remains to be established. More specifically, researchers continue to suggest that the SPOS yields scores with high reliability (e.g., Rhoades & Eisenberger, 2002).

Reliability Generalization

Reliability is an issue bound in scores obtained from a sample of individuals responding to a given measure of interest rather than to a particular test (Thompson, 2003). Samples, sampling procedures, and testing situations, among other unique study characteristics, can all have systematic effects on measurement error such that induced comparisons between any new study and previous empirical results may not be appropriate. The psychometric literature is clear that reliability is grounded in scores obtained from a given measurement scale and will vary from sample to sample (Henson, 2001; Vacha-Haase, Henson, & Caruso, 2002). The American Psychological Association (APA) Task Force on Statistical Inference (Wilkinson & APA Task Force, 1999), the fifth edition of the APA's *Publication Manual* (APA, 2001), the *Standards for Educational and Psychological Testing* (American Educational Research Association, APA, & National Council on Measurement in Education [AERA, APA, & NCME], 1999), and some editorial policies (e.g., Thompson, 2003) remind researchers that tests, in and among themselves, are not reliable or unreliable; rather, scores obtained from samples using a particular scale, with a given number of items, under different circumstances have varying reliability properties and therefore should be reported in *all* studies and not simply induced. However, some researchers continue to induce score reliability based upon previous empirical research even when samples may not be comparable and differing numbers of items are in use (Yoon & Lin, 1999). Reliability induction occurs when the reliability score characteristics obtained from previous studies are assumed to be similar to those obtained from new samples, or worse, when researchers believe that reliability is a function of the test itself and does not need to be recomputed (cf. Vacha-Haase et al., 2002; Vacha-Haase, Kogan, & Thompson, 2000). Systematic sources of measurement error serve to reduce effect size estimates and statistical power (Henson, 2001; Pedhazur, 1997). When these sources of error are not considered by the researcher in each new sample, they can have consequences in interpretation relative to observed effect sizes and their confidence intervals (Charter & Feldt, 2002).

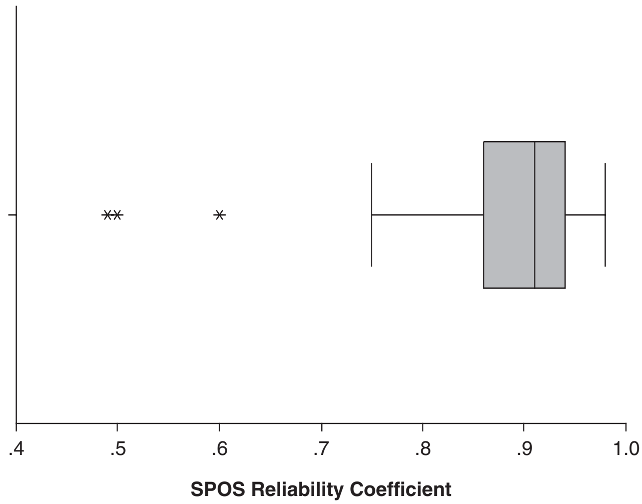
RG reflects a meta-analytic approach to quantify reliability scores across samples and examine the variability of these estimates (Vacha-Haase, 1998). Reliability coefficients are collected across studies to calculate central tendency and variability tendencies. Sample characteristics can be tabulated in an attempt to account for the variability in reliability scores when used as a dependent variable (e.g., Yin & Fan, 2000). At its most basic level, RG can provide evidence regarding the mean reliability (and variability) for POS scores when samples are weighted across studies. More complex RG analyses (e.g., general linear models) can provide guidelines when the scale's use is more or less favorable. For instance, distributions of age, gender, supervisory level, type of organization, sample size, and number of items can be subjected to ANOVA or regression models to add clarity to our understanding of the variability among POS reliability estimates.

Method: Data and Procedure

The sources for reliability coefficients were those obtained from more than 70 studies used in a meta-analysis by Rhoades and Eisenberger (2002) concerning the POS. A total of 48 studies that were reviewed by Rhoades and Eisenberger were included in this study. More specifically, we did not include the dissertations, theses, and unpublished manuscripts and were unable to locate 6 of the studies that were included in Rhoades and Eisenberger. In addition, a search of leading journals in organizational development, management, and industrial/organizational psychology produced an additional 14 studies with reported reliability coefficients for the SPOS. Therefore, a total of 62 studies were used in the present analyses.

One concern highlighted by previous RG studies (Thompson, 2003) is the number of primary studies that induce score reliability from other sources. Each study was classified into one of two groups. The first group included studies in which the authors induced reliability of the SPOS scores, either through citing specific coefficients from other studies, referencing other studies that had reported reliability although not providing specific coefficients, or implying the reliability of the instrument was well established without providing specific citations (n = 2, 3.2%). The second group of studies included those in which the authors reported reliability coefficients computed on the data analyzed in the article (n = 56, 90.3%). Four studies did not report a reliability coefficient (6.5%). When authors reported multiple reliability coefficients for the data at hand (e.g., separate coefficients reported for subgroups of participants, or independent samples for a separate study reported within the same published article), each reported coefficient was used. Specifically, there were 8 published articles in which multiple reliability coefficients were reported for separate subgroups of participants, producing 27 independent observations. All reported reliability coefficients were internal consistency and did not include test-retest scores.

Figure 1
Boxplot of Survey of Perceived Organizational Support (SPOS) Score Reliabilities (*n* = 58)



For each study, sample characteristics and the reliability coefficients themselves were recorded following suggestions by Henson and Thompson (2002). Study features and sample characteristics available for coding included (a) mean age of the study participants in years, (b) mean tenure of study participants in years, (c) percentage female of participants in each study (d) percentage minority of participants in each study, (e) sample size used in each study, and (f) number of SPOS items used in each study.

Results

The mean reliability coefficient for the 58 studies reporting internal consistency was .88 (*SD* = .10; *SE* = .134) with a 95% confidence interval ranging from .851 to .904 (cf. Henson & Thompson, 2002). The median reliability coefficient was .90 (mode = .94) with scores ranging from .49 to .98. The distribution was negatively skewed (–2.311; *SE* = 0.314) with kurtosis = 5.902 (*SE* = 0.618). This distribution quality will be discussed further as it may have important methodological implications for RG studies as a whole. This distribution of SPOS reliability scores is further illustrated in Figure 1.

Table 1
Zero-Order Correlation Matrix of Survey of Perceived Organizational Support (SPOS) Reliability Scores and Coded Study Characteristics

	M	*SD*	1	2	3	4	5	6	7
1. SPOS α	0.88	0.10	—						
2. Number of SPOS items	11.64	5.68	.60	—					
3. Sample size	282.09	240.72	–.10	–.01	—				
4. Mean age	39.32	5.93	.57	.48	–.29	—			
5. Mean tenure	9.02	7.81	.19	.19	–.30	.76	—		
6. Percentage female	41.11	18.76	.10	.10	–.05	–.36	–.11	—	
7. Percentage minority	16.89	8.25	–.34	–.26	.20	–.31	.25	.17	—

Each variable coded for the studies was examined individually in relation to the reliability coefficients reported for the POS instrument. In these initial analyses, three variables were found to have statistically significant correlations with the reliability coefficients. Sample size initially correlated with the reliability coefficients at $r_{xy} = -.28$, $p = .034$. However, an extreme outlier on the sample size variable was found with $N = 2{,}136$ ($z = 5.50$). This case also had a large Cook's *D* value (1.79). Thus, the decision was made to eliminate this extreme case, which yielded a statistically nonsignificant relationship, $r_{xy} = -.10$, $p = .489$.

Table 1 provides the correlation matrix for the variables included in this study along with descriptives. The correlation between mean age of samples and the reliability coefficients was also found to be statistically significant, $r_{xy} = .57$, $p < .001$. This analysis included 33 studies that had provided both mean age for the samples and the reliability coefficient. Two studies were deleted because of outlying lower reliability coefficients equal to .49 and .50. Due to our interest in the relationship of mean ages to reliability coefficients, residual scores (standardized, studentized, deleted) for these outlying cases were generated. These two lowest coefficients (.49 and .50) were found to have residual values of –2.81 and –3.43, respectively. The decision was made based on the extreme nature of these values to eliminate these two cases from all analyses.

Examination of the scatter plot between mean ages of samples with the reliability coefficients depicted a noticeable trend for higher reliabilities at the higher levels of mean ages (constant = .661; $\beta = .57$). This trend was even more dramatic before the outliers were removed. The studies reporting reliabilities less than .80 were in the lower mean age range.

The number of items included on the instrument also yielded a statistically significant correlation, $r_{xy} = .60$, $p < .001$. The tendency for reliability to increase as the number of items increased was expected and is well established (cf. Cortina, 1993; Crocker & Algina, 1986; Nunnally & Bernstein, 1994). For the studies included in this analysis, the number of items used ranged from a low of 3 ($n = 3$) to a high of 36 ($n = 1$). The SPOS shorter versions of 8 ($n = 11$) and 16 ($n = 5$) were also used as were several other item totals. Indeed, the mean number of SPOS items used was 11.64 ($SD = 5.68$), with the median items as 10.00 and the mode 17.00.

Whereas it is not surprising that the number of items used was positively and strongly correlated with the reliability coefficients, we also examined whether the mean age of the sample predicted variance in the reliability coefficients beyond the number of items. Using reliability estimates as the dependent variable, we computed a regression analysis with the mean age of sample added as a second predictor to the number of items on the scale entered as a first predictor. The R^2 value with number of SPOS items at the first step was .36, and it increased to .47 when the mean age of the sample was added, a statistically significant increment, $F(1, 30) = 6.173$, $p < .001$. However, the increment also reflects a statistically significant correlation between the two predictors, $r_{xy} = .48$, $p < .01$, which may be meaningful in further examination of the relations.

Discussion

The primary goal of the current research was to employ RG techniques to investigate the magnitude and variability of reliability estimates obtained across studies using the SPOS developed by Eisenberger et al. (1986). As seen in the results, the average score reliability was high with some, albeit minimal, variability. Our study shows a negative skew in the distribution of reliability coefficients, which is expected as researchers strive to find measures with minimal measurement error. However, as RG studies employ the general linear model to explain variability in reliability coefficients, decisions will need to consider logarithmically transforming these data to reflect normal distributions.

A second goal for this research was to investigate factors that might influence the score reliability obtained using the SPOS. Specifically, this study examined such study characteristics as number of items, sample size, percentage female, percentage minority, and industry type as potential artifacts that might account for reliability score variability. Initially, our study found three variables having statistically significant correlations with reliability scores. However, after eliminating extreme outliers, only number of items used and mean age remained as significantly (and substantially) correlated with reliability scores obtained from studies using various forms of the SPOS.

Given that SPOS scores are argued to represent a unidimensional construct, researchers may be seduced into using fewer SPOS items for economical reasons. Rhoades and Eisenberger (2002) stated, "The original scale is unidimensional and has high internal consistency, the use of shorter versions does not appear problematic" (p. 699). However, given our results, we suggest some caution when making the decision to use an extreme reduction in the number of SPOS items as reliability may be reduced, thereby masking subsequent effect size indicators especially if the average intercorrelations between the items selected are low (Cortina, 1993). Indeed, alpha is a function of the average interitem correlation and the number of items (Henson, 2001; Nunnally & Bernstein, 1994). Determining the number of items to use in the measure of POS involves the assumption that the measure, regardless of how many items, demonstrates identical validities to other measures of interest to the researcher. In practice, the researcher would need to pay particular attention to sampling of item content concerns between a longer and shorter measure. Furthermore, it is assumed that the various number of items used to measure POS are randomly drawn from the same universe of item correlations. When sampling from a universe with higher interitem correlations (more homogeneous content), fewer items are needed to maximize alpha. If the universe is more heterogeneous in nature, more items are needed to adequately represent POS. In our data, 4 studies used three SPOS items, resulting in an average score reliability of .71. Applying the Spearman Brown prophecy formula, increasing the number of items from three to eight results in an estimated alpha score of .93. To that end, 11 studies in our analyses used eight SPOS items with an average alpha score of .90. Given the influence of score reliabilities on effect size estimate (cf. Wilkinson & APA Task Force, 1999), the use of few items to measure POS is clearly a decision that should be made after careful thought.

Mean age was shown to also have a strong positive correlation with reliability scores obtained using the SPOS. The scatter plot between mean age and reliability illustrated that as mean age increased among the studies, a corresponding increase in reliability scores with reduced variability was also observed. Given these findings, we computed a hierarchical regression equation on score reliabilities to estimate the amount of variance that mean age would account beyond that accounted for by the number of items used. Mean age did account for a statistically significant amount of variance in score reliability over and above the number of items. This presents a new advancement by which to consider organizational support theory in general and the measurement of POS in particular. However, prior to discussing our findings with respect to the influence of mean age on reliability scores obtained using the SPOS, the reader is reminded to consider ecological fallacies with regard to implications of our findings (cf. Pedhazur, 1997; Robinson, 1950). That is, the observed correlation obtained from the current study is based upon the aggregate of mean age and reliability scores across samples and may not hold at the individual unit of analysis. Nevertheless, previous research has demonstrated a positive relationship between age and various work attitudes such as job satisfaction, job involvement, and organizational commitment (cf. Rhodes, 1983).

The possible influence of age on the measurement of POS warrants further attention. Cursory examination of the SPOS items developed by Eisenberger et al. (1986) suggests that several items may be related to age. For example, being replaced by someone new at a lower salary, organizational understanding of long absences due to illness and employment for the remainder of one's career may be perceived differently by older employees compared to younger newcomers. It is important to note that for the purposes of this study, average tenure did not correlate with reliability scores, $r_{xy} = .19$, $p = .273$; although average tenure was strongly correlated with mean age, $r_{xy} = .76$, $p < .001$. To the extent that future research discovers statistically significant and meaningful item-level correlations between age and the SPOS items, organizational support theorists will need to specifically address age and researchers will need to carefully consider the items they include in their measurement of POS. For instance, it would be interesting to examine structural differences between older and younger employees relative to the dimensionality of the SPOS across its different forms.

In this study, the average score reliability and relatively small standard deviation suggests psychometric success across samples. With a growing focus by professional organizations and journal editors to present effect size indices (cf. Wilkinson & APA Task Force, 1999), it is imperative that researchers also report observed score reliability to assist our ability to evaluate the meaningfulness of effect sizes within the presence of measurement error. Nevertheless, some researchers continue to induce score reliability across disparate samples and items. It is encouraging that less than 5% of the studies used in the present research actually induced score reliability.

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